Statistical Thermodynamics

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Statistical Thermodynamics allows us to describe the thermodynamic behavior of a large collection of atoms or molecules ($\sim 10^{23}$) under equilibrium conditions, given the interaction between the atoms / molecules.

Macrostates and microstates

We know that the thermodynamic state of any system can be specified by two independent properties and masses of each component in the system.
e.g. for a pure system, we can specify the thermodynamic state with N (number of molecules), V (volume) and T (temperature)
We call this as the "macrostate" of the system.

We call also specify the state of the system through the positions $(\vec{r}_1, \ldots, \vec{r}_N)$ and momenta $(\vec{p}_1, \ldots, \vec{p}_N)$ of all atoms.
We call this as the "microstate" of the system.

For a given macrostate, there exists a collection of microstates.
This collection is also known as an "ensemble" of microstates.

A property that depends on the microstate is called "mechanical property".
Since every thermodynamic state is a collection of microstates, the value of properties that one measures in an experiment is a time averaged property.

Since every microstate has a certain probability of occurrence, we can also calculate an ensemble average of the property.

Two postulates of Statistical Thermodynamics

1) The time average value of any mechanical property is equal to its ensemble average.

Let $M(\vec{r}^N, \vec{p}^N)$ be any mechanical property.

$$\text{Time average} = \langle M \rangle_t = \lim_{T \to \infty} \int_0^T M \, dt$$

$$\text{Ensemble average} = \langle M \rangle_e = \int \cdots \int M(\vec{r}^N, \vec{p}^N) \, f(\vec{r}^N, \vec{p}^N) \, d\vec{r}^N \, d\vec{p}^N$$

↑ probability density

According to 1st postulate

$$\langle M \rangle_t = \langle M \rangle_e$$

Such systems are called "ergodic" systems.

2) All micro-states that have same energy have same probability of occurence.

Types of ensembles

1) **Micro-canonical ensemble**

This ensemble represents an isolated system

The thermodynamic state is specified by
- N : number of molecules
- V : volume
- E : Total energy.

Since energy is constant, all microstates in the N, VE ensemble have same probability of occurrence.

Let Ω be the total number of microstates

Probability of a microstate $\equiv Pr\{\nu\} = \frac{1}{\Omega}$ $\quad \nu \equiv$ microstate

$\Omega \equiv$ partition function of the NVE ensemble
$\equiv \Omega(N, V, E)$

- Entropy $S = k_B \ln \Omega$

2) **Canonical Ensemble**

This represents a closed system at constant volume.

Thermodynamic state is specified by N, V, and T

we can show that $Pr\{\nu\} \propto e^{-\beta E_\nu}$

$$\beta \equiv \frac{1}{k_B T}, \quad k_B \equiv \text{Boltzmann's constant} = 1.38 \times 10^{-23} \frac{J}{K}$$

Partition function $Q = \sum_\nu e^{-\beta E_\nu}$

$$\therefore Pr\{\nu\} = \frac{e^{-\beta E_\nu}}{Q}$$

Helmholtz free energy $A = -k_B T \ln Q$

For classical systems

Probability density $f(\vec{r}^N, \vec{p}^N) \propto e^{-\beta \mathcal{H}(\vec{r}^N, \vec{p}^N)}$

$\mathcal{H} \equiv$ Hamiltonian $\equiv$ Potential energy + Kinetic energy

$$Q = \frac{1}{h^{3N} N!} \underbrace{\int_0^L \cdots \int_0^L \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty}}_{6N \text{ dimensional integral}} e^{-\beta \mathcal{H}} \, d\vec{r}^N \, d\vec{p}^N$$

$d\vec{r}_i \equiv dx_i \, dy_i \, dz_i$

$d\vec{p}_i = dp_{x_i} \, dp_{y_i} \, dp_{z_i}$

$$f(\vec{r}^N, \vec{p}^N) = \frac{e^{-\beta \mathcal{H}}}{\int \cdots \int e^{-\beta \mathcal{H}} \, d\vec{r}^N \, d\vec{p}^N}$$

It is possible to integrate the kinetic energy

$$f(\vec{r}^N) = \frac{e^{-\beta U(\vec{r}^N)}}{\underbrace{\int \cdots \int e^{-\beta U} \, d\vec{r}^N}_{\text{configurational integral}}}$$

$U \equiv$ potential energy

Ensemble average $\langle M \rangle_e = \int \cdots \int M(\vec{r}^N, \vec{p}^N) \, f(\vec{r}^N, \vec{p}^N) \, d\vec{r}^N \, d\vec{p}^N$

$$= \int \cdots \int M(\vec{r}^N, \vec{p}^N) \frac{e^{-\beta \mathcal{H}} \, d\vec{r}^N \, d\vec{p}^N}{\int \cdots \int e^{-\beta \mathcal{H}} \, d\vec{r}^N \, d\vec{p}^N}$$

$$= \frac{\int \cdots \int M(\vec{r}^N, \vec{p}^N) \, e^{-\beta \mathcal{H}} \, d\vec{r}^N \, d\vec{p}^N}{\int \cdots \int e^{-\beta \mathcal{H}} \, d\vec{r}^N \, d\vec{p}^N}$$

3) Grand canonical ensemble

This represents an open system at constant volume.

Thermodynamic state is specified by μ, V, T (μ: chemical potential)

Most commonly applied to study adsorption.

$$\text{Prob}_j \propto e^{-\beta E_j} e^{\beta \mu N_j}$$

Partition function $\Xi(\mu, V, T) = \sum_{N=0}^{\infty} e^{\beta \mu N} Q(N, V, T)$

$$= \sum_{N=0}^{\infty} \sum_{E} e^{\beta\mu N} e^{-\beta E} \Omega(N,V,E)$$

$$= \sum_{\nu} e^{\beta\mu N_\nu} e^{-\beta E_\nu}$$

Grand potential $= PV = k_B T \ln \Xi$

For classical systems

$$\Xi(\mu,V,T) = \sum_{N=0}^{\infty} \frac{e^{\beta\mu N}}{h^{3N} N!} \int\cdots\int e^{-\beta\mathcal{H}} d\vec{r}^{N} d\vec{p}^{N}$$

$$\langle M \rangle = \frac{\sum_{N=0}^{\infty} \frac{e^{\beta\mu N}}{h^{3N} N!} \int\cdots\int M\, e^{-\beta\mathcal{H}} d\vec{r}^{N} d\vec{p}^{N}}{\sum_{N=0}^{\infty} \frac{e^{\beta\mu N}}{h^{3N} N!} \int\cdots\int e^{-\beta\mathcal{H}} d\vec{r}^{N} d\vec{p}^{N}}$$

4) Isothermal - isobaric ensemble

This represents closed systems at constant pressure

Thermodynamic state is specified by N, P, T.

$$P_\nu\{\nu\} \propto e^{-\beta P V_\nu} e^{-\beta E_\nu}$$

Partition function $= \Delta(N,P,T) = \sum_{\nu} e^{-\beta E_\nu} e^{-\beta P V_\nu}$

For classical systems

$$\Delta(N,P,T) = \frac{C}{N!} \int_0^{\infty} e^{-\beta P V} \int\cdots\int e^{-\beta\mathcal{H}} d\vec{r}^{N} d\vec{p}^{N} \, dV$$

Gibbs free energy $= G = -k_B T \ln \Delta$